

## SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

### ASSESSMENT TOOL 1 – CONCEPT MAP

|                  |                                                                                                                       |               |                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------|
| Course Code:     | ITA03                                                                                                                 | Course Title: | Mobile computing                |
| CO Assessed:     | CO1                                                                                                                   | Assessment:   | Assessment Tool 1 – Concept Map |
| Weightage:       | 60 % of CIA                                                                                                           | Total Marks:  | 20 marks                        |
| CO1 Statement:   | Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems. |               |                                 |
| Student Name:    | K V CHARAN TEJA                                                                                                       | Reg. No.:     | 192424167                       |
| Section / Batch: | C                                                                                                                     | Date:         | 10/10/2026                      |

#### Code of Conduct

*I **K V CHARAN TEJA(192424167)** (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student:*

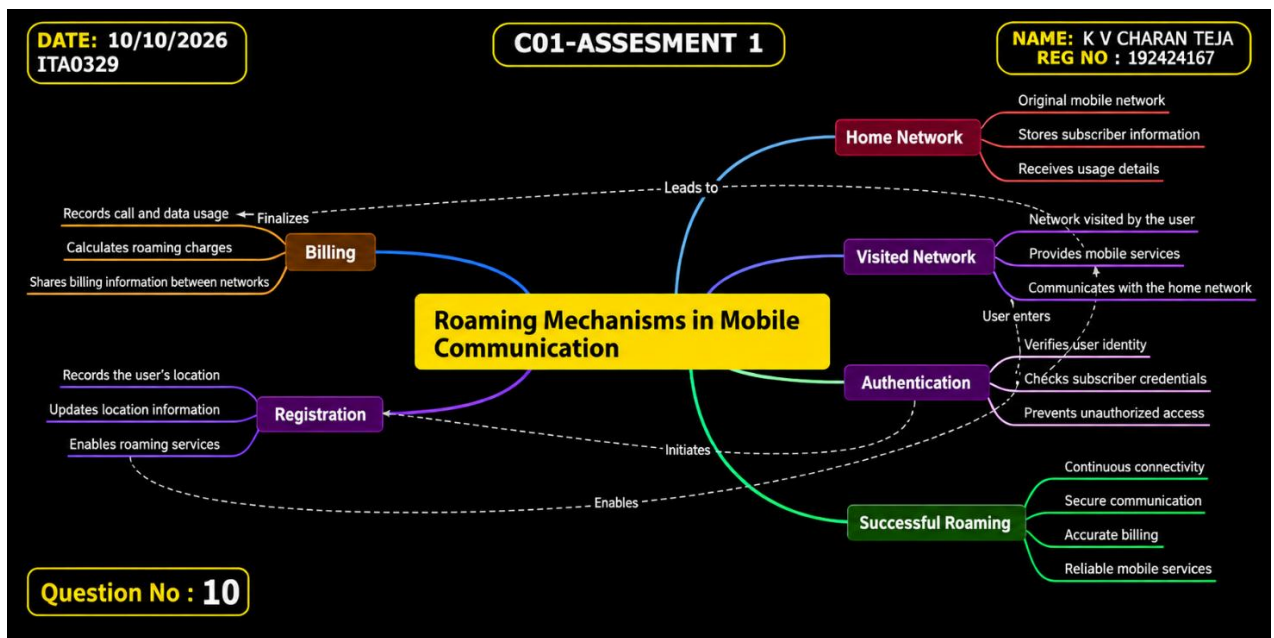
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#### Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

## Annexure A - QUESTIONS

10. Construct a concept map that evaluates the effectiveness of roaming mechanisms in mobile communication.
- Illustrate relationships among home network, visited network, authentication, registration, and billing.
- Assess the importance of each process in successful roaming.
- Support your conclusions through logical concept organization.



### Rubrics for Calculation

| Criteria                         | Weightage | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    |
|----------------------------------|-----------|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20        | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts          | 30        | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |
| Problem Solving Approach         | 20        | Logical, well structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |
| Accuracy of Solution             | 20        | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |
| Clarity                          | 10        | Clear, well organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |